AWS Cloud Practical

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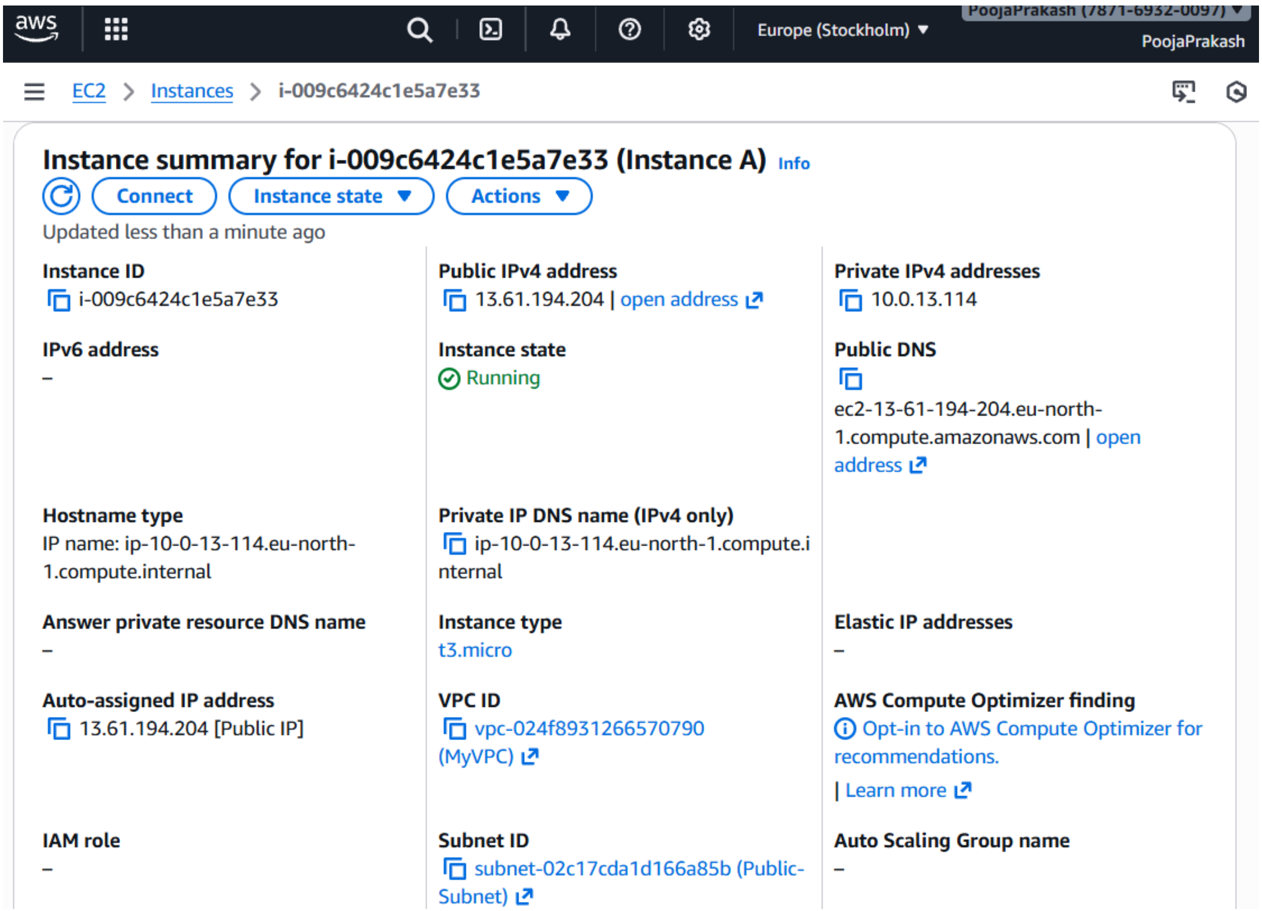
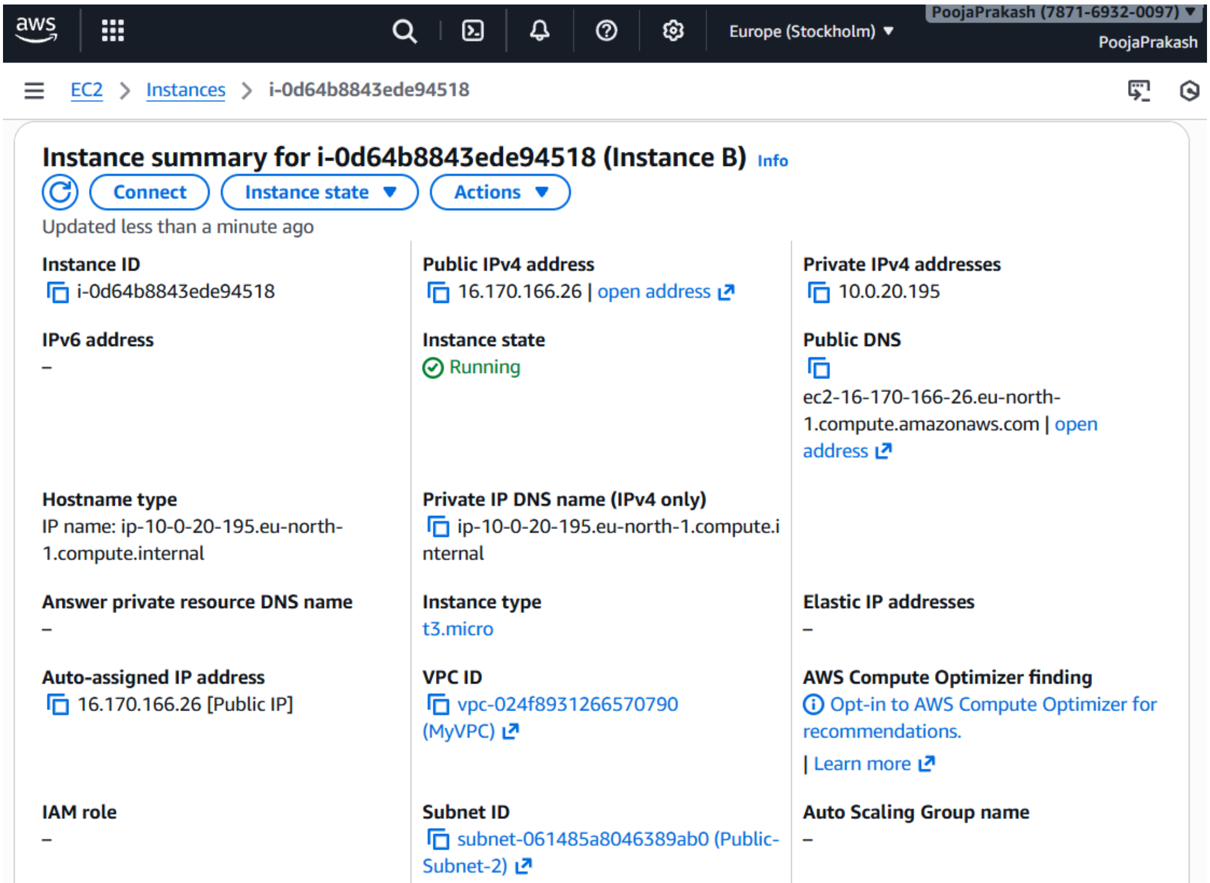
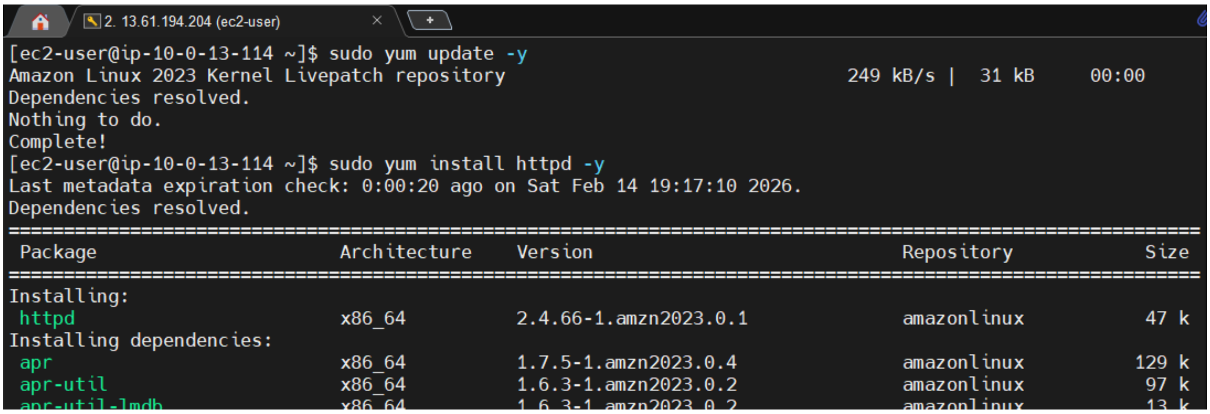
1. Deploy a static website on two EC2 instances in different Availability Zones and configure an Application Load Balancer (ALB) to distribute traffic between them to achieve high availability.

- Launch two EC2 instances named Instance A and Instance B in the same VPC but in different Availability Zones (AZs) using public subnets. Allow inbound HTTP traffic on port 80.
- Different AZs are used to achieve high availability. If one AZ fails, traffic can still be served from the other AZ.





- Install the Apache web server (httpd) on both EC2 instances, then start and enable the service.



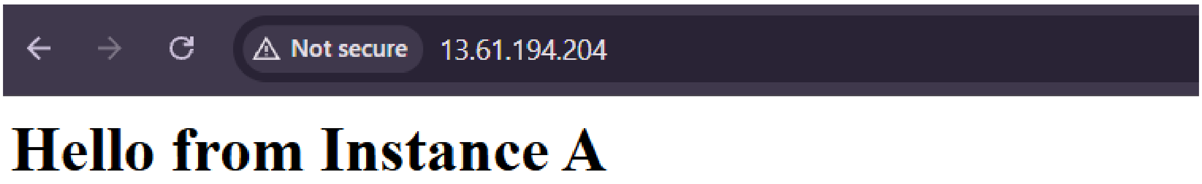
```
[ec2-user@ip-10-0-13-114 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-13-114 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.se
rvice.
[ec2-user@ip-10-0-13-114 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-02-14 19:17:47 UTC; 27s ago
       Docs: man:httpd.service(8)
   Main PID: 25837 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1067)
     Memory: 13.3M
        CPU: 77ms
     CGroup: /system.slice/httpd.service
             ├─25837 /usr/sbin/httpd -DFOREGROUND
             ├─25838 /usr/sbin/httpd -DFOREGROUND
             ├─25839 /usr/sbin/httpd -DFOREGROUND
             ├─25840 /usr/sbin/httpd -DFOREGROUND
             └─25841 /usr/sbin/httpd -DFOREGROUND

Feb 14 19:17:47 ip-10-0-13-114.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache >
Feb 14 19:17:47 ip-10-0-13-114.eu-north-1.compute.internal systemd[1]: Started httpd.service - The Apache H>
Feb 14 19:17:47 ip-10-0-13-114.eu-north-1.compute.internal httpd[25837]: Server configured, listening on: p>

[ec2-user@ip-10-0-13-114 ~]$
```

- Create a webpage named index.html inside the /var/www/html directory on both instances.
  Add the content:
  - "Hello from Instance A" on Instance A
  - "Hello from Instance B" on Instance B

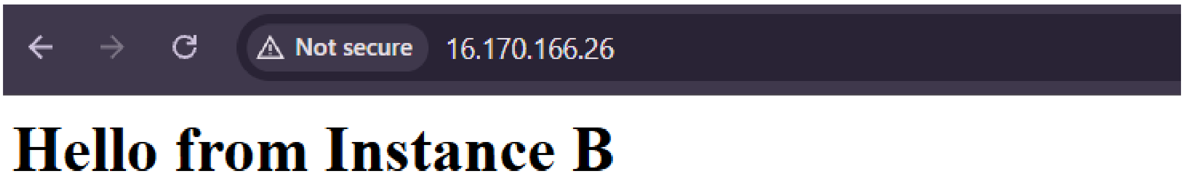
```
2. 13.61.194.204 (ec2-user)
  GNU nano 8.3                                  /var/www/html/index.html
<h1>Hello from Instance A</h1>
```

```
2. Instance A    3. Instance B
  GNU nano 8.3                                  /var/www/html/index.html
<h1>Hello from Instance B</h1>
```

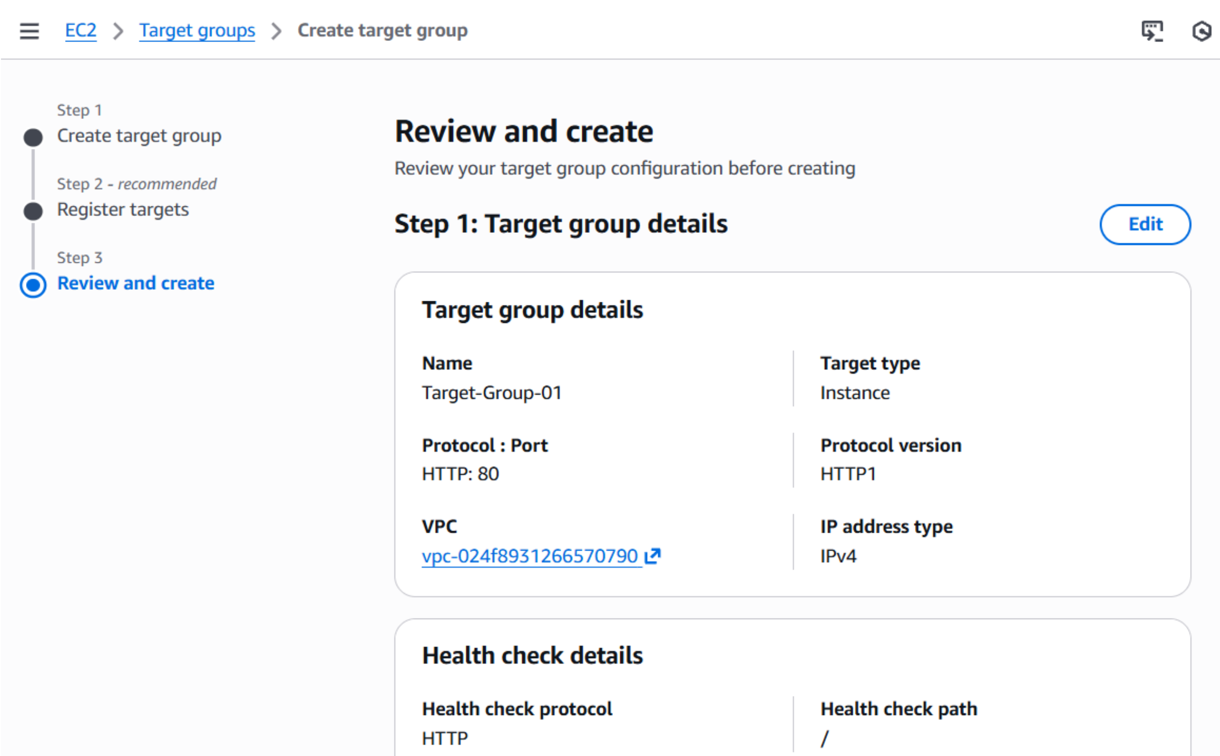
- Verify that both webpages are accessible using the public IP addresses of the respective instances.
    - Webpage of Instance A

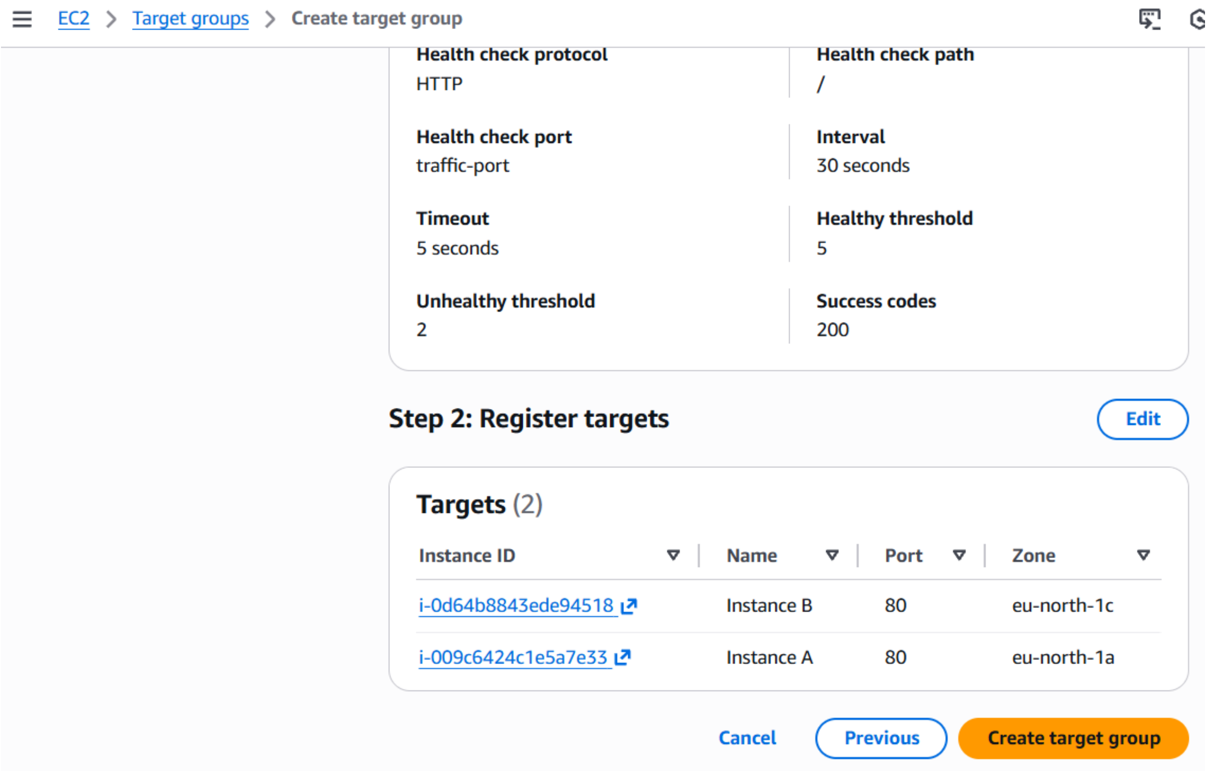


    - Webpage of Instance B

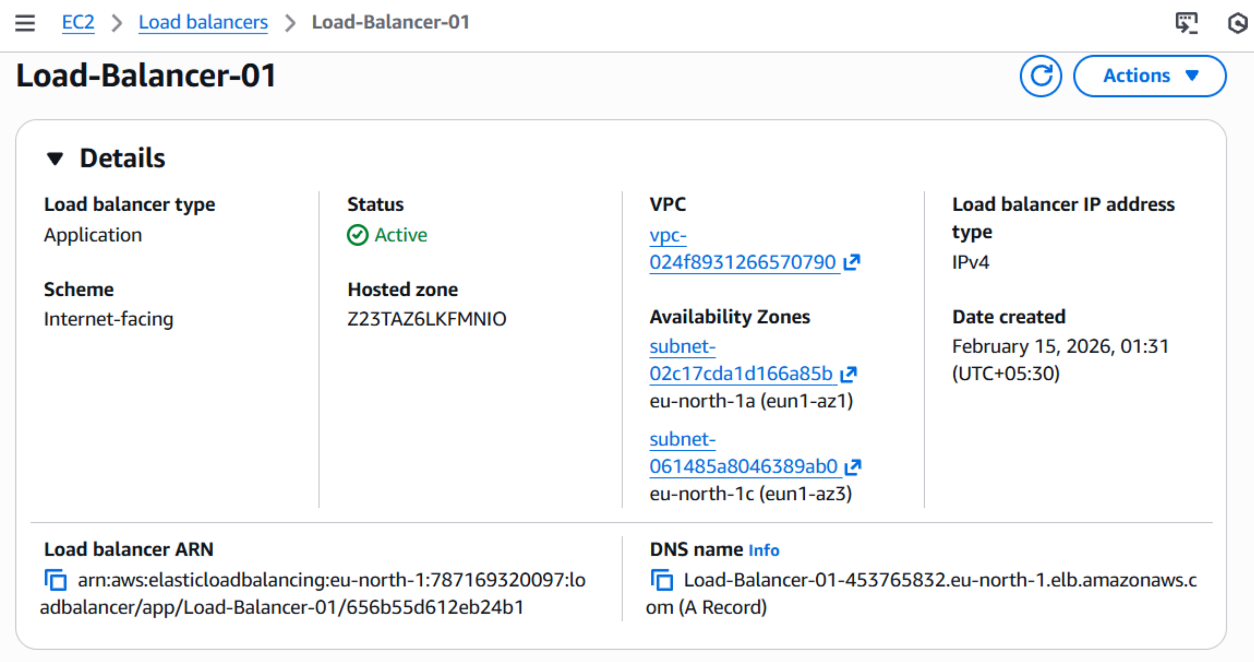


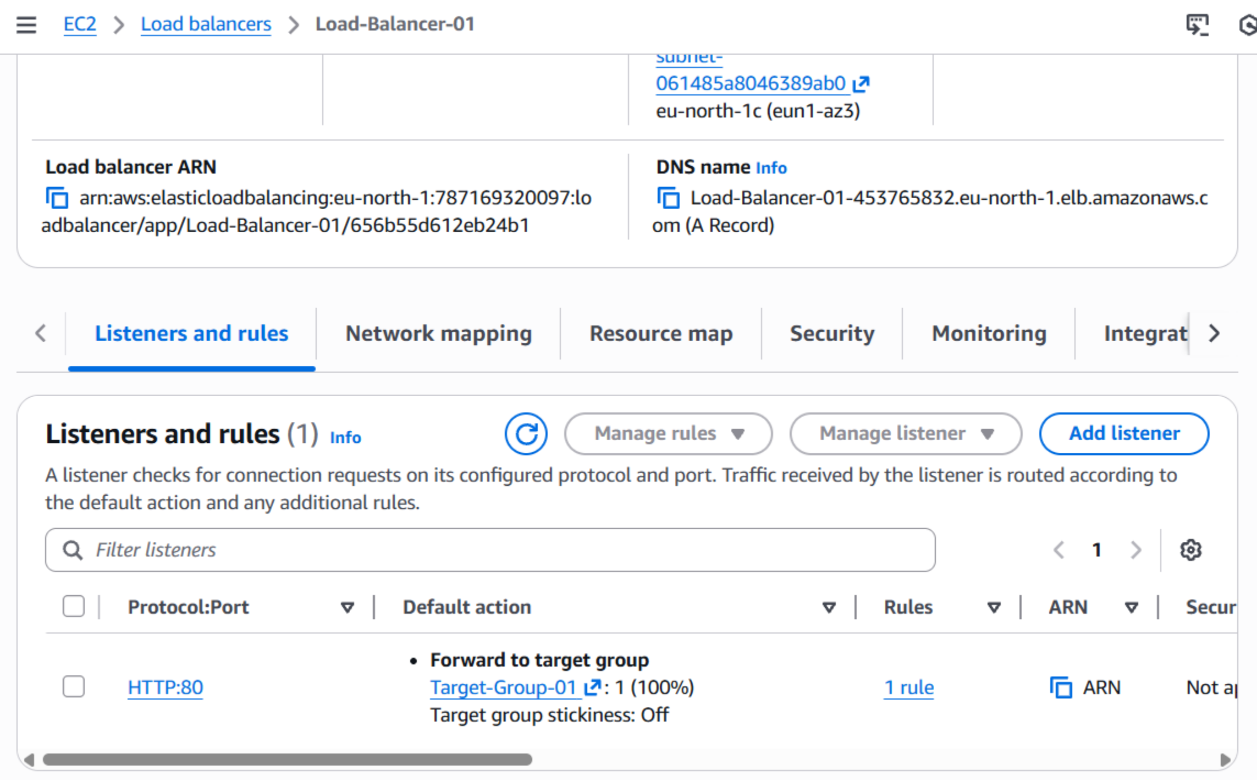
- Create a Target Group in the same VPC as the EC2 instances. Set the Health Check path to / and register both Instance A and Instance B as targets.





- Create an Application Load Balancer (ALB) as Internet-facing in the same VPC. Select two subnets that are located in different Availability Zones.
- Attach a Security Group to the ALB that allows HTTP (port 80) from 0.0.0.0/0. Configure the Listener on HTTP port 80 and forward traffic to the created Target Group.


EC2 > Load balancers > Load-Balancer-01
Load-Balancer-01
Actions
Details
Load balancer type
Application
Status
Active
VPC
vpc-024f8931266570790
Load balancer IP address type
IPv4
Scheme
Internet-facing
Hosted zone
Z23TAZ6LKFMNIO
Availability Zones
subnet-02c17cda1d166a85b
eu-north-1a (eun1-az1)
subnet-061485a8046389ab0
eu-north-1c (eun1-az3)
Date created
February 15, 2026, 01:31 (UTC+05:30)
Load balancer ARN
arn:aws:elasticloadbalancing:eu-north-1:787169320097:loadbalancer/app/Load-Balancer-01/656b55d612eb24b1
DNS name Info
Load-Balancer-01-453765832.eu-north-1.elb.amazonaws.com (A Record)


EC2 > Load balancers > Load-Balancer-01
subnet-061485a8046389ab0
eu-north-1c (eun1-az3)
Load balancer ARN
arn:aws:elasticloadbalancing:eu-north-1:787169320097:loadbalancer/app/Load-Balancer-01/656b55d612eb24b1
DNS name Info
Load-Balancer-01-453765832.eu-north-1.elb.amazonaws.com (A Record)
Listeners and rules
Network mapping
Resource map
Security
Monitoring
Integrat
Listeners and rules (1) Info
Manage rules
Manage listener
Add listener
A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.
Filter listeners
1
Protocol:Port
Default action
Rules
ARN
Secur
HTTP:80
Forward to target group
Target-Group-01: 1 (100%)
Target group stickiness: Off
1 rule
ARN
Not a

- Access the application using the Load Balancer's DNS name. Refresh the page multiple times to observe traffic being distributed between Instance A and Instance B.